TECHNOLOGY STACK

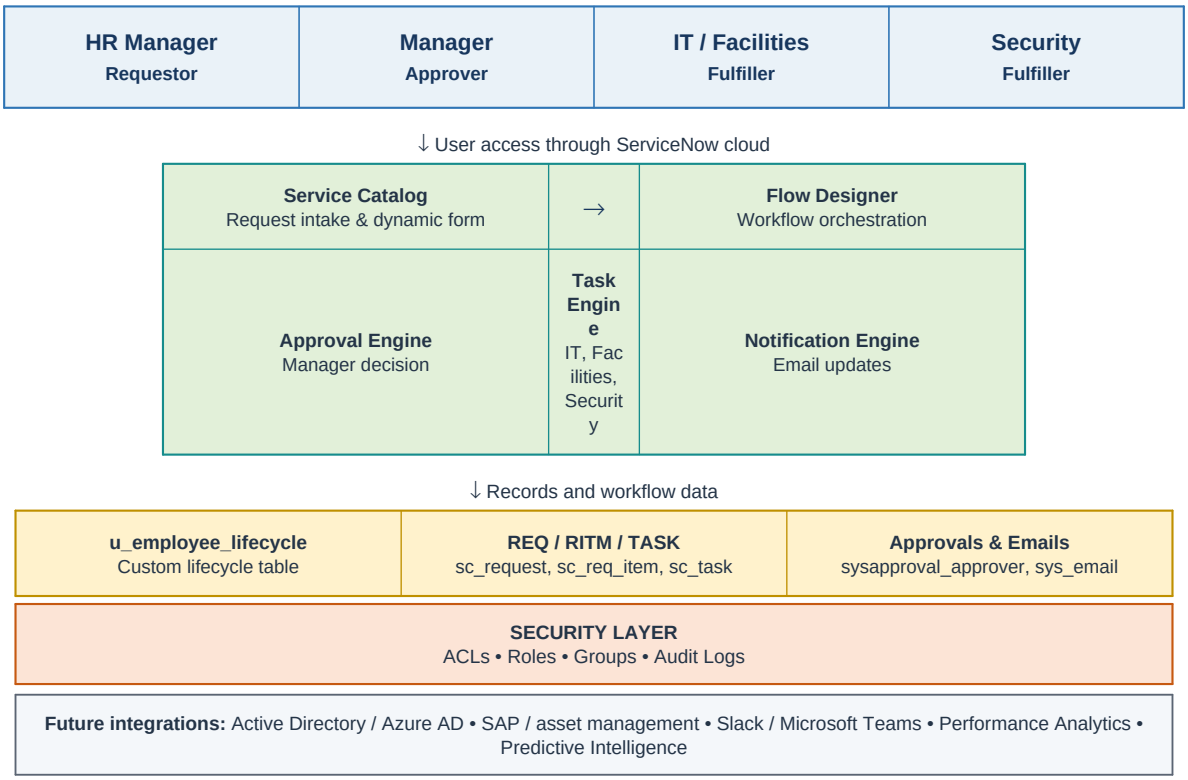
Architecture & Stack | Project Design Phase-II

Date	15 October 2026	**Team ID**	SkillWallet — ServiceNow Project
Project Name	Automated Employee Onboarding & Offboarding System	**Maximum Marks**	4 Marks

1. Technical Architecture

The solution uses a three-tier architecture on the ServiceNow cloud platform. Users submit onboarding or offboarding requests through the Service Catalog; Flow Designer orchestrates manager approvals, departmental tasks, notifications, and lifecycle updates. Security controls apply across the application and data layers.

Architecture Diagram



Note: External integrations and predictive features are marked as future scope in the supplied project content.

2. Table 1 — Technology Stack Components

S.No	Component	Description	Technology
1	User Interface	Service Portal for requestors; ServiceNow agent UI for approvers and fulfillers	HTML5, CSS3, ServiceNow Service Portal, AngularJS (ServiceNow UI Framework)
2	Application Logic-1	Intake — dynamic catalog form with 13 variables and UI policies	ServiceNow Service Catalog (no-code form builder)
3	Application Logic-2	Orchestration — approvals, task creation, status updates, notifications	ServiceNow Flow Designer (15 actions, no custom code)
4	Application Logic-3	Manager approval routing based on the catalog Manager field	ServiceNow Approval Engine + sysapproval_approver table
5	Application Logic-4	Creates and assigns three catalog tasks to IT, Facilities, and Security	ServiceNow Catalog Task Engine (sc_task table)
6	Application Logic-5	Emails on approval, completion, and rejection	ServiceNow Notification Engine + SMTP
7	Database (Custom)	Stores onboarding/offboarding lifecycle data	ServiceNow MySQL-based table u_employee_lifecycle
8	Cloud Database	Platform-hosted database for REQ, RITM, TASK, and custom records	ServiceNow Cloud Database (MySQL variant)
9	File Storage	Attachments on catalog requests, including documents and images	ServiceNow Attachment Storage (cloud)
10	External API-1 (future)	Automated employee account and email provisioning	Microsoft Active Directory / Azure AD via Integration Hub Spoke
11	External API-2 (future)	Real-time notifications to teams	Slack / Microsoft Teams via Integration Hub Spoke
12	External API-3 (future)	Asset management system handoff	SAP / Dell / HP Asset Management API
13	Machine Learning Model (future)	Predictive SLA breach warnings and smart task assignment	ServiceNow Predictive Intelligence
14	Infrastructure (Server / Cloud)	Application deployed on ServiceNow cloud (PaaS)	ServiceNow Cloud (Personal Developer Instance for dev/test)

3. Table 2 — Application Characteristics

S.No	Characteristic	Description	Technology
1	Open-Source Frameworks	ServiceNow UI uses AngularJS for Service Portal widgets. REST APIs are planned for future integrations.	AngularJS 1.x (ServiceNow UI framework), JavaScript (minimal client scripts for UI policies)
2	Security Implementations	Role-based access through ACLs (read/write/create/delete); HR, IT, Facilities, and Security groups mapped to roles; admin override for admins; access logged.	ServiceNow ACLs, Roles, Groups, SSO-ready, encryption at rest and in transit (TLS 1.2+)
3	Scalable Architecture	Three-tier architecture: Presentation (Portal) → Application (Flow Designer) → Data (Tables). Content states Flow Designer supports concurrent requests without degradation.	ServiceNow PaaS — auto-scaling cloud platform; horizontal scalability
4	Availability	The supplied project content states 99.9%+ uptime, no single point of failure, and global load balancing.	ServiceNow Cloud SLA, multi-tenant cloud infrastructure
5	Performance	The supplied project content states flow execution under 2 seconds per trigger (1347 ms runtime), portal form load under 3 seconds, and second-accurate SLA timers.	ServiceNow in-memory processing, indexing on key table fields
6	Maintainability	100% low-code; no Business Rules or Script Includes. Admin can modify flows visually; documentation provided for replication.	Flow Designer, Admin UI, Markdown documentation

S.No	Characteristic	Description	Technology
7	Auditability	Audit trail includes flow execution details, approval history, task state changes, and email logs.	ServiceNow Audit tables, Flow Execution Contexts

4. Architectural Demarcation

Layer	Local / Cloud	Details
User Layer	Local (user device)	Browser-based access to Service Portal
Application Layer	Cloud (ServiceNow)	Flow Designer, Catalog, Approval Engine
Data Layer	Cloud (ServiceNow)	All tables hosted on ServiceNow cloud database
External Interfaces	Cloud (future)	Active Directory, asset management, Slack via Integration Hub
Machine Learning	Cloud (future)	ServiceNow Predictive Intelligence

5. Data Storage Summary

Storage	Type	Purpose
u_employee_lifecycle	Custom table	Onboarding/offboarding records
sc_request	Out-of-box table	Top-level request (REQ)
sc_req_item	Out-of-box table	Requested item (RITM)
sc_task	Out-of-box table	Departmental catalog tasks
sysapproval_approver	Out-of-box table	Approval records
sys_email	Out-of-box table	Email notification logs
sys_hub_flow_context	Out-of-box table	Flow execution history
Attachments	Cloud storage	Files uploaded with requests

6. External Interfaces — Future Scope

Interface	Purpose	Technology
Active Directory	Auto-provision user accounts	Integration Hub AD Spoke
Azure AD	SSO and group synchronization	OAuth 2.0 / SAML
Slack / Teams	Real-time approval notifications	Integration Hub Slack/Teams Spoke
SAP / Asset Management	Hardware procurement	REST API via Integration Hub
Performance Analytics	Executive dashboards	ServiceNow Performance Analytics (built-in)

7. Infrastructure Summary

ServiceNow Cloud (PaaS)
Multi-tenant cloud platform • Auto-scaling on demand • 99.9%+ uptime SLA • Global data centres • Automated backups • Built-in disaster recovery
Development / Test: Personal Developer Instance (PDI) | **Production:** ServiceNow enterprise cloud

8. Summary

Aspect	Value
Architecture Type	Three-tier (Presentation → Application → Data)
Deployment	Cloud (ServiceNow PaaS)
Frontend	Service Portal (AngularJS-based)
Backend	Flow Designer (visual, low-code)
Database	ServiceNow MySQL-based cloud database
Security	ACLs, Roles, Groups, Audit Logs
Scalability	Horizontal (auto-scaling cloud)
Availability	99.9%+ uptime (as stated in project content)
Custom Scripts	0 (100% low-code, as stated in project content)
Future Integrations	Active Directory, Slack, SAP, Predictive Intelligence

References

• C4 Model: c4model.com
• IBM Architecture: ibm.com/cloud/architecture
• AWS Architecture: aws.amazon.com/architecture
• IBM online order processing pattern and the architecture-diagram guidance listed in the project template.